A digital twin the jury can challenge.

Choose. Freeze. Reveal. Compare.

An interactive test of the forecast, its failure boundaries and the artifacts behind the result.

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v2 supplement · unchanged model · generated data only · not clinical validation

Freeze the available past. Reveal the recorded future.

The freeze response omits future values; the public dataset makes this a demonstration protocol.

01

Choose

Patient, cutoff and sensor condition. Copy and truncate history.

02

Freeze

Infer on the prefix. Preserve output, quality and fingerprints.

03

Reveal

Compare the frozen result with original recorded future.

04

Export

Download the experiment receipt.

Outcome table

Forecast error · persistence error · interval hit · exact +120-minute label
No recalculation on reveal. A selected window is not a cohort metric.

Four cases. Four different lessons.

The reference future is unchanged; each challenge modifies only its input copy.

Clean replay

Original available history.
Inspect an honest baseline.

10-minute CGM gap

Last two glucose samples removed.
Degraded data; forward filling.

30-minute dropout

Last six glucose samples removed.
Stale input; prediction withheld.

Undetected sensor bias

+30 mg/dL over the recent CGM window.
Can pass freshness checks while wrong.

Quality checks are not a certificate of sensor correctness.

Changed routines expose clock dependence.

Frozen model · fixed 0.20 alert threshold · 48 held-out patients / 6,720 windows per case

Meal timing	MAE	Persistence	Coverage	Recall
Original schedule	10.95	22.10	90.0%	83.9%
Delayed +2 hours	21.62	21.66	61.7%	28.7%
Delayed +4 hours	**22.78**	**19.99**	**61.3%**	**21.7%**
Irregular (1.5h SD)	16.60	21.64	76.5%	57.3%

MAE in mg/dL at +120 min; interval coverage is for nominal 90% bands.

The +4h case loses to persistence. This audit exposes a limitation; it does not fix it.

A receipt for the experiment, not a trust certificate.

SHA-256 fingerprints identify content. They do not establish clinical validity or trusted authorship.

01

Inside the receipt

Patient + cutoff + case
Frozen prediction and quality
Model version and content fingerprint
Perturbed input-prefix fingerprint
Revealed outcome, when available

02

Explicit boundaries

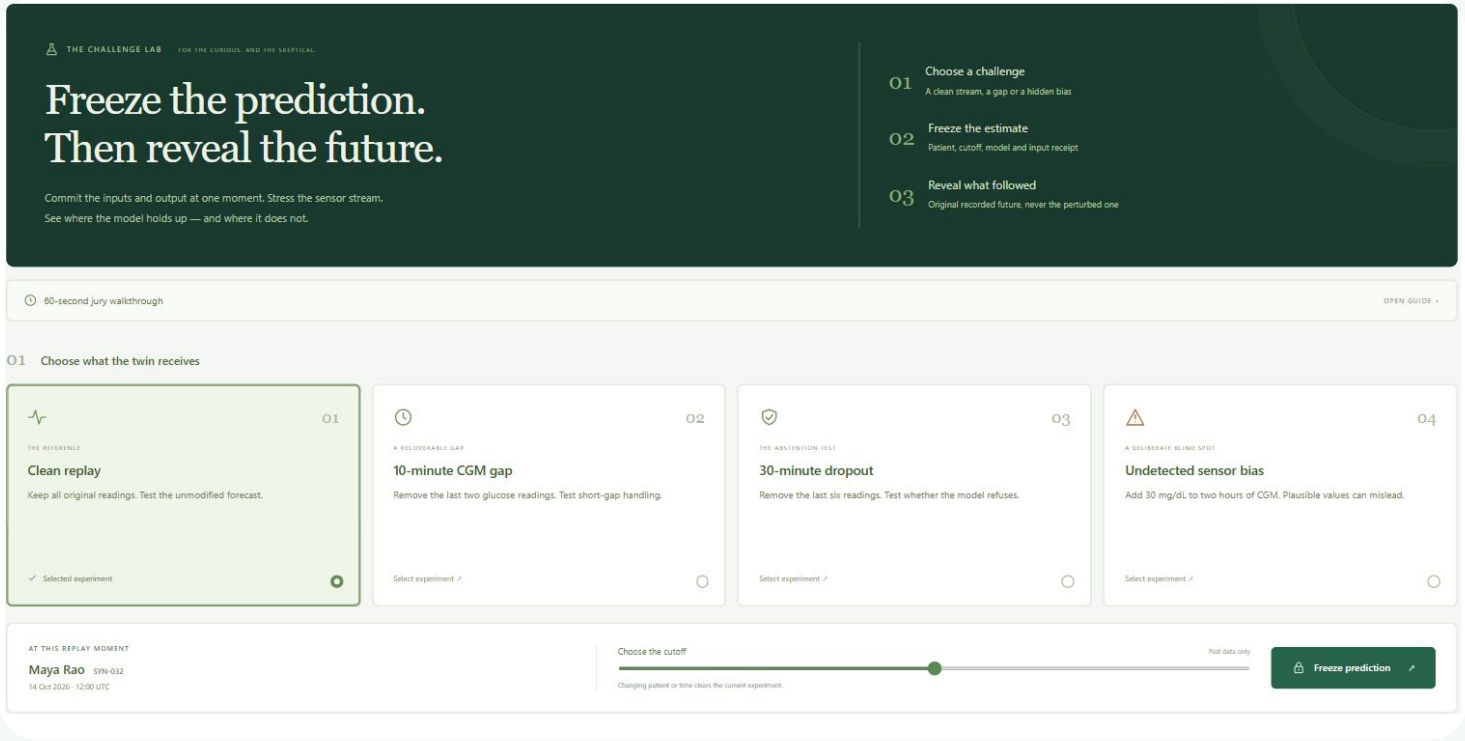
Unsigned; no independent timestamp
Memory-only experiment sessions
One-hour expiry; 128-session cap
No clinical identity or audit governance
No claim of tamper-proof evidence

Offline check: node scripts/verify_receipt.mjs receipt.json

Recomputes hashes, prediction and reveal. Consistency does not authenticate the author or timestamp.

A three-minute challenge. A reviewable result.

Let the jury choose a supported patient and cutoff, then inspect the result together.



Our defensible distinction

A working, inspectable product-and-evidence loop.

No “first glucose twin” claim.
No clinical efficacy claim.
No causal treatment claim.

Related work: GlyTwin (arXiv:2504.09846) · VVUQ perspective (npj Digital Medicine, 2025)